

TP 01 — Practical Work

Python: What It Is and How It Speaks

8 exercises • no answers in this sheet

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October 2026

How to work through this TP

- Do the exercises **in order**. Each one uses only what Chapter 01 and the chapters before it have taught — nothing else is needed, and nothing else is expected.
- **Type the code yourself**. Do not copy and paste. Typing is what builds the memory, and you cannot fix an error you did not write.
- Run your file and compare your output with what the exercise asks for. If they differ, read the error message from the bottom up: the last line names the real mistake.
- Only then open the companion sheet *Solutions — Chapter 01*. Reading the answer before trying teaches you nothing.

□ **What to hand in** – One Python file per exercise that produces code, named `ex1.py` to `ex8.py`, plus a short written answer for the exercises that ask you to explain rather than to compute.

The 8 exercises

Exercise 1. The timeline (written answer) – Without looking back at the chapter, write the year of each of these five events: (a) Guido van Rossum starts Python; (b) the first public release; (c) Python 3.0 appears; (d) Python 2 stops being supported; (e) the version you should install today, and its number.

Exercise 2. Why Python, and the glue (written answer) – (a) Give four reasons why Python is used so widely, and for each one a single sentence on why it matters to a FinTech student in particular. (b) Explain in one sentence what a *glue language* is, then explain why calling Python slow is only half true.

Exercise 3. Your first program (code) – Write a file that prints three lines: the text **Banking fee report**, the year 2026 as a number, and the VAT of 19% on a fee of 1 250 DZD. Put a comment above the third line saying which rate is applied.

Hint: text goes in quotes, numbers do not; # starts a comment.

Exercise 4. Variables and an f-string (code) – A sales agent is paid a commission of 2.5% on what he sells. Last month he sold 1 340 000 DZD. Store the sales figure and the commission rate in two named variables, compute the commission in a third, then

print one line reading 1340000 DZD of sales earns 33500.00 DZD commission — the amount shown with exactly two decimals.

Hint: inside an f-string, {value:.2f} shows two decimals.

Exercise 5. Two kinds of division (code) – An invoice of 7 450 DZD is to be split between 4 people. Display, on three lines: the exact share, the whole number of dinars each person pays, and the remainder left over. Then verify the identity $a = b \times (a // b) + (a \% b)$ by printing its right-hand side.

Hint: the three division operators are /, // and %.

Exercise 6. Powers and rounding (code) – A bank card PIN is 4 digits, each one of 0–9. The matching online account uses a password of 8 characters drawn from the 62 letters and digits. Display, on three lines: (a) the number of possible PINs; (b) the number of possible 8-character passwords; (c) if a machine can test one PIN every 0.35 seconds, the number of hours needed to try them all, rounded to 2 decimals.

Hint: n choices repeated k times gives n^k ; the operator is `**`.

Exercise 7. Types and conversion (code) – Print the type of each of these four values: 250000, 0.035, "250000" and `7 / 1`. Then, in two further lines, add 8 to the text "42" to obtain 50, and join the text "Total: " to the number 1250.

Hint: `type()` names a type; `int()` and `str()` convert.

Exercise 8. Read the error (written answer) – (a) A program fails with `NameError: name 'capitl' is not defined` at line 3. What is wrong, and how do you fix it? (b) A tutorial contains the line `print "Total: ", total`. What is wrong with it, what is the correct Python 3 line, and what does it tell you about the tutorial? (c) A user types 250000 into `input()` and the program then computes `capital * 0.035`. It fails with a `TypeError`. Why, and what single change repairs it?